

VOLUME I: FOUNDATIONS

The Quantum Theory of Fields

Steven Weinberg

University of Texas at Austin

To Louise

Contents

Sections marked with an asterisk are somewhat out of the book's main line of development and may be omitted in a first reading.

PREFACE	xx
NOTATION	xxv
1 HISTORICAL INTRODUCTION	1
1.1 Relativistic Wave Mechanics	3
De Broglie waves • Schrödinger–Klein–Gordon wave equation • Fine structure • Spin • Dirac equation • Negative energies • Exclusion principle • Positrons • Dirac equation reconsidered	
1.2 The Birth of Quantum Field Theory	15
Born, Heisenberg, Jordan quantized field • Spontaneous emission • Anticommutators • Heisenberg–Pauli quantum field theory • Furry–Oppenheimer quantization of Dirac field • Pauli–Weisskopf quantization of scalar field • Early calculations in quantum electrodynamics • Neutrons • Mesons	
1.3 The Problem of Infinities	31
Infinite electron energy shifts • Vacuum polarization • Scattering of light by light • Infrared divergences • Search for alternatives • Renormalization • Shelter Island Conference • Lamb shift • Anomalous electron magnetic moment • Schwinger, Tomonaga, Feynman, Dyson formalisms • Why not earlier?	
Bibliography	39
References	40
2 RELATIVISTIC QUANTUM MECHANICS	49
2.1 Quantum Mechanics	49
Rays • Scalar products • Observables • Probabilities	
2.2 Symmetries	50
Wigner's theorem • Antilinear and antiunitary operators • Observables • Group structure • Representations up to a phase • Superselection rules • Lie groups • Structure constants • Abelian symmetries	
2.3 Quantum Lorentz Transformations	55
Lorentz transformations • Quantum operators • Inversions	
2.4 The Poincaré Algebra	58
$J^{\mu\nu}$ and P^μ • Transformation properties • Commutation relations • Conserved and non-conserved generators • Finite translations and rotations • Inönü–Wigner contraction • Galilean algebra	
2.5 One-Particle States	62
Transformation rules • Boosts • Little groups • Normalization • Massive particles • Massless particles • Helicity and polarization	
2.6 Space Inversion and Time-Reversal	74
Transformation of $J^{\mu\nu}$ and P^μ • P is unitary and T is antiunitary • Massive particles • Massless particles • Kramers degeneracy • Electric dipole moments	
2.7 Projective Representations*	81
Two-cocycles • Central charges • Simply connected groups • No central charges in the Lorentz group • Double connectivity of the Lorentz group • Covering groups • Superselection rules reconsidered	
Appendix A The Symmetry Representation Theorem	91
Appendix B Group Operators and Homotopy Classes	96
Appendix C Inversions and Degenerate Multiplets	100
Problems	104
References	105

3	SCATTERING THEORY	107
3.1	‘In’ and ‘Out’ States	107
	Multi-particle states • Wave packets • Asymptotic conditions at early and late times • Lippmann–Schwinger equations • Principal value and delta functions	
3.2	The S-Matrix	113
	Definition of the S -matrix • The T -matrix • Born approximation • Unitarity of the S -matrix	
3.3	Symmetries of the S-Matrix	116
	Lorentz invariance • Sufficient conditions • Internal symmetries • Electric charge, strangeness, isospin, $SU(3)$ • Parity conservation • Intrinsic parities • Pion parity • Parity non-conservation • Time-reversal invariance • Watson’s theorem • PT non-conservation • C , CP , CPT • Neutral K -mesons • CP non-conservation	
3.4	Rates and Cross-Sections	134
	Rates in a box • Decay rates • Cross-sections • Lorentz invariance • Phase space • Dalitz plots	
3.5	Perturbation Theory	141
	Old-fashioned perturbation theory • Time-dependent perturbation theory • Time-ordered products • The Dyson series • Lorentz-invariant theories • Distorted wave Born approximation	
3.6	Implications of Unitarity	147
	Optical theorem • Diffraction peaks • CPT relations • Particle and antiparticle decay rates • Kinetic theory • Boltzmann H -theorem	
3.7	Partial-Wave Expansions*	151
	Discrete basis • Expansion in spherical harmonics • Total elastic and inelastic cross-sections • Phase shifts • Threshold behavior: exothermic, endothermic, and elastic reactions • Scattering length • High-energy elastic and inelastic scattering	
3.8	Resonances*	159
	Reasons for resonances: weak coupling, barriers, complexity • Energy-dependence • Unitarity • Breit–Wigner formula • Unresolved resonances • Phase shifts at resonance • Ramsauer–Townsend effect	
	Problems	165
	References	166
4	THE CLUSTER DECOMPOSITION PRINCIPLE	169
4.1	Bosons and Fermions	170
	Permutation phases • Bose and Fermi statistics • Normalization for identical particles	
4.2	Creation and Annihilation Operators	173
	Creation operators • Calculating the adjoint • Derivation of commutation/anticommutation relations • Representation of general operators • Free-particle Hamiltonian • Lorentz transformation of creation and annihilation operators • C , P , T properties of creation and annihilation operators	
4.3	Cluster Decomposition and Connected Amplitudes	177
	Decorrelation of distant experiments • Connected amplitudes • Counting delta functions	
4.4	Structure of the Interaction	182
	Condition for cluster decomposition • Graphical analysis • Two-body scattering implies three-body scattering	
	Problems	189
	References	189

5	QUANTUM FIELDS AND ANTIPARTICLES	191
5.1	Free Fields	191
	Creation and annihilation fields • Lorentz transformation of the coefficient functions • Construction of the coefficient functions • Implementing cluster decomposition • Lorentz invariance requires causality • Causality requires antiparticles • Field equations • Normal ordering	
5.2	Causal Scalar Fields	201
	Creation and annihilation fields • Satisfying causality • Scalar fields describe bosons • Antiparticles • P, C, T transformations • π^0	
5.3	Causal Vector Fields	207
	Creation and annihilation fields • Spin zero or spin one • Vector fields describe bosons • Polarization vectors • Satisfying causality • Antiparticles • Mass zero limit • P, C, T transformations	
5.4	The Dirac Formalism	213
	Clifford representations of the Poincaré algebra • Transformation of Dirac matrices • Dimensionality of Dirac matrices • Explicit matrices • γ_5 • Pseudounitariness • Complex conjugate and transpose	
5.5	Causal Dirac Fields	219
	Creation and annihilation fields • Dirac spinors • Satisfying causality • Dirac fields describe fermions • Antiparticles • Space inversion • Intrinsic parity of particle–antiparticle pairs • Charge-conjugation • Intrinsic C-phase of particle–antiparticle pairs • Majorana fermions • Time-reversal • Bilinear covariants • Beta decay interactions	
5.6	General Irreducible Representations of the Homogeneous Lorentz Group*	229
	Isomorphism with $SU(2) \otimes SU(2)$ • (A, B) representation of familiar fields • Rarita–Schwinger field • Space inversion	
5.7	General Causal Fields*	233
	Constructing the coefficient functions • Scalar Hamiltonian densities • Satisfying causality • Antiparticles • General spin-statistics connection • Equivalence of different field types • Space inversion • Intrinsic parity of general particle–antiparticle pairs • Charge-conjugation • Intrinsic C-phase of antiparticles • Self-charge-conjugate particles and reality relations • Time-reversal • Problems for higher spin?	
5.8	The CPT Theorem	244
	CPT transformation of scalar, vector, and Dirac fields • CPT transformation of scalar interaction density • CPT transformation of general irreducible fields • CPT invariance of Hamiltonian	
5.9	Massless Particle Fields	246
	Constructing the coefficient functions • No vector fields for helicity ± 1 • Need for gauge invariance • Antisymmetric tensor fields for helicity ± 1 • Sums over helicity • Constructing causal fields for helicity ± 1 • Gravitons • Spin ≥ 3 • General irreducible massless fields • Unique helicity for (A, B) fields	
	Problems	255
	References	256
6	THE FEYNMAN RULES	259
6.1	Derivation of the Rules	259
	Pairings • Wick’s theorem • Coordinate space rules • Combinatoric factors • Sign factors • Examples	
6.2	Calculation of the Propagator	274
	Numerator polynomial • Feynman propagator for scalar fields • Dirac fields • General irreducible fields • Covariant propagators • Non-covariant terms in time-ordered products	
6.3	Momentum Space Rules	280
	Conversion to momentum space • Feynman rules • Counting independent momenta • Examples • Loop suppression factors	

6.4	Off the Mass Shell	286
	Currents • Off-shell amplitudes are exact matrix elements of Heisenberg-picture operators • Proof of the theorem	
	Problems	290
	References	291
7	THE CANONICAL FORMALISM	292
7.1	Canonical Variables	293
	Canonical commutation relations • Examples: real scalars, complex scalars, vector fields, Dirac fields • Free-particle Hamiltonians • Free-field Lagrangian • Canonical formalism for interacting fields	
7.2	The Lagrangian Formalism	298
	Lagrangian equations of motion • Action • Lagrangian density • Euler–Lagrange equations • Reality of the action • From Lagrangians to Hamiltonians • Scalar fields revisited • From Heisenberg to interaction picture • Auxiliary fields • Integrating by parts in the action	
7.3	Global Symmetries	306
	Noether’s theorem • Explicit formula for conserved quantities • Explicit formula for conserved currents • Quantum symmetry generators • Energy–momentum tensor • Momentum • Internal symmetries • Current commutation relations	
7.4	Lorentz Invariance	314
	Currents $\mathcal{M}^{\rho\mu\nu}$ • Generators $J^{\mu\nu}$ • Belinfante tensor • Lorentz invariance of S -matrix	
7.5	Transition to Interaction Picture: Examples	318
	Scalar field with derivative coupling • Vector field • Dirac field	
7.6	Constraints and Dirac Brackets	325
	Primary and secondary constraints • Poisson brackets • First and second class constraints • Dirac brackets • Example: real vector field	
7.7	Field Redefinitions and Redundant Couplings*	331
	Redundant parameters • Field redefinitions • Example: real scalar field	
	Appendix Dirac Brackets from Canonical Commutators	332
	Problems	337
	References	338
8	ELECTRODYNAMICS	339
8.1	Gauge Invariance	339
	Need for coupling to conserved current • Charge operator • Local symmetry • Photon action • Field equations • Gauge-invariant derivatives	
8.2	Constraints and Gauge Conditions	343
	Primary and secondary constraints • Constraints are first class • Gauge fixing • Coulomb gauge • Solution for A^0	
8.3	Quantization in Coulomb Gauge	346
	Remaining constraints are second class • Calculation of Dirac brackets in Coulomb gauge • Construction of Hamiltonian • Coulomb interaction	
8.4	Electrodynamics in the Interaction Picture	350
	Free-field and interaction Hamiltonians • Interaction picture operators • Normal mode decomposition	
8.5	The Photon Propagator	353
	Numerator polynomial • Separation of non-covariant terms • Cancellation of non-covariant terms	

8.6	Feynman Rules for Spinor Electrodynamics	355
	Feynman graphs • Vertices • External lines • Internal lines • Expansion in $\alpha/4\pi$ • Circular, linear, and elliptic polarization • Polarization and spin sums	
8.7	Compton Scattering	362
	S -matrix • Differential cross-section • Kinematics • Spin sums • Traces • Klein–Nishina formula • Polarization by Thomson scattering • Total cross-section	
8.8	Generalization: p-Form Gauge Fields*	369
	Motivation • p -forms • Exterior derivatives • Closed and exact p -forms • p -form gauge fields • Dual fields and currents in D spacetime dimensions • p -form gauge fields equivalent to $(D - p - 2)$ -form gauge fields • Nothing new in four spacetime dimensions	
	Appendix Traces	372
	Problems	374
	References	375
9	PATH-INTEGRAL METHODS	376
9.1	The General Path-Integral Formula	378
	Transition amplitudes for infinitesimal intervals • Transition amplitudes for finite intervals • Interpolating functions • Matrix elements of time-ordered products • Equations of motion	
9.2	Transition to the S-Matrix	385
	Wave function of vacuum • $i\epsilon$ terms	
9.3	Lagrangian Version of the Path-Integral Formula	389
	Integrating out the ‘momenta’ • Derivatively coupled scalars • Non-linear sigma model • Vector field	
9.4	Path-Integral Derivation of Feynman Rules	395
	Separation of free-field action • Gaussian integration • Propagators: scalar fields, vector fields, derivative coupling	
9.5	Path Integrals for Fermions	399
	Anticommuting c-numbers • Eigenvectors of canonical operators • Summing states by Berezin integration • Changes of variables • Transition amplitudes for infinitesimal intervals • Transition amplitudes for finite intervals • Derivation of Feynman rules • Fermion propagator • Vacuum amplitudes as determinants	
9.6	Path-Integral Formulation of Quantum Electrodynamics	413
	Path integral in Coulomb gauge • Reintroduction of a^0 • Transition to covariant gauges	
9.7	Varieties of Statistics*	418
	Preparing ‘in’ and ‘out’ states • Composition rules • Only bosons and fermions in ≥ 3 dimensions • Anyons in two dimensions	
	Appendix Gaussian Multiple Integrals	420
	Problems	423
	References	423
10	NON-PERTURBATIVE METHODS	425
10.1	Symmetries	425
	Translations • Charge conservation • Furry’s theorem	
10.2	Polology	428
	Pole formula for general amplitudes • Derivation of the pole formula • Pion exchange	
10.3	Field and Mass Renormalization	436
	LSZ reduction formula • Renormalized fields • Propagator poles • No radiative corrections in external lines • Counterterms in self-energy parts	

10.4	Renormalized Charge and Ward Identities	442
	Charge operator • Electromagnetic field renormalization • Charge renormalization • Ward–Takahashi identity • Ward identity	
10.5	Gauge Invariance	448
	Transversality of multi-photon amplitudes • Schwinger terms • Gauge terms in photon propagator • Structure of photon propagator • Zero photon renormalized mass • Calculation of Z_3 • Radiative corrections to choice of gauge	
10.6	Electromagnetic Form Factors and Magnetic Moment	452
	Matrix elements of J^0 • Form factors of J^μ : spin zero • Form factors of J^μ : spin $\frac{1}{2}$ • Magnetic moment of a spin $\frac{1}{2}$ particle • Measuring the form factors	
10.7	The Källén–Lehmann Representation*	457
	Spectral functions • Causality relations • Spectral representation • Asymptotic behavior of propagators • Poles • Bound on field renormalization constant • $Z = 0$ for composite particles	
10.8	Dispersion Relations*	462
	History • Analytic properties of massless boson forward scattering amplitude • Subtractions • Dispersion relation • Crossing symmetry • Pomernichuk’s theorem • Regge asymptotic behavior • Photon scattering	
	Problems	469
	References	470
11	ONE-LOOP RADIATIVE CORRECTIONS IN QUANTUM ELECTRODYNAMICS	472
11.1	Counterterms	472
	Field, charge, and mass renormalization • Lagrangian counterterms	
11.2	Vacuum Polarization	473
	One-loop integral for photon self-energy part • Feynman parameters • Wick rotation • Dimensional regularization • Gauge invariance • Calculation of Z_3 • Cancellation of divergences • Vacuum polarization in charged particle scattering • Uehling effect • Muonic atoms	
11.3	Anomalous Magnetic Moments and Charge Radii	485
	One-loop formula for vertex function • Calculation of form factors • Anomalous lepton magnetic moments to order α • Anomalous muon magnetic moment to order $\alpha^2 \ln(m_\mu/m_e)$ • Charge radius of leptons	
11.4	Electron Self-Energy	493
	One-loop formula for electron self-energy part • Electron mass renormalization • Cancellation of ultraviolet divergences	
	Appendix Assorted Integrals	497
	Problems	498
	References	498
12	GENERAL RENORMALIZATION THEORY	499
12.1	Degrees of Divergence	500
	Superficial degree of divergence • Dimensional analysis • Renormalizability • Criterion for actual convergence	
12.2	Cancellation of Divergences	505
	Subtraction by differentiation • Renormalization program • Renormalizable theories • Example: quantum electrodynamics • Overlapping divergences • BPHZ renormalization prescription • Changing the renormalization point; ϕ^4 theory	

12.3	Is Renormalizability Necessary?	516
	Renormalizable interactions cataloged • No renormalizable theories of gravitation • Cancellation of divergences in non-renormalizable theories • Suppression of non-renormalizable interactions • Limits on new mass scales • Problems with higher derivatives? • Detection of non-renormalizable interactions • Low-energy expansions in non-renormalizable theories • Example: scalar with only derivative coupling • Saturation or new physics? • Effective field theories	
12.4	The Floating Cutoff*	525
	Wilson's approach • Renormalization group equation • Polchinski's theorem • Attraction to a stable surface • Floating cutoff vs renormalization	
12.5	Accidental Symmetries*	529
	General renormalizable theory of charged leptons • Redefinition of the lepton fields • Accidental conservation of lepton flavors, P, C, and T	
	Problems	531
	References	532
13	INFRARED EFFECTS	534
13.1	Soft Photon Amplitudes	534
	Single photon emission • Negligible emission from internal lines • Lorentz invariance implies charge conservation • Single graviton emission • Lorentz invariance implies equivalence principle • Multi-photon emission • Factorization	
13.2	Virtual Soft Photons	539
	Effect of soft virtual photons • Radiative corrections on internal lines	
13.3	Real Soft Photons; Cancellation of Divergences	544
	Sum over helicities • Integration over energies • Sum over photon number • Cancellation of infrared cutoff factors • Likewise for gravitation	
13.4	General Infrared Divergences	548
	Massless charged particles • Infrared divergences in general • Jets • Lee–Nauenberg theorem	
13.5	Soft Photon Scattering*	553
	Poles in the amplitude • Conservation relations • Universality of the low-energy limit	
13.6	The External Field Approximation*	556
	Sums over photon vertex permutations • Non-relativistic limit • Crossed ladder exchange	
	Problems	562
	References	562
14	BOUND STATES IN EXTERNAL FIELDS	564
14.1	The Dirac Equation	565
	Dirac wave functions as field matrix elements • Anticommutators and completeness • Energy eigenstates • Negative energy wave functions • Orthonormalization • 'Large' and 'small' components • Parity • Spin- and angle-dependence • Radial wave equations • Energies • Fine structure • Non-relativistic approximations	
14.2	Radiative Corrections in External Fields	572
	Electron propagator in an external field • Inhomogeneous Dirac equation • Effects of radiative corrections • Energy shifts	
14.3	The Lamb Shift in Light Atoms	578
	Separating high and low energies • High-energy term • Low-energy term • Effect of mass renormalization • Total energy shift • $\ell = 0$ • $\ell \neq 0$ • Numerical results • Theory vs experiment for classic Lamb shift • Theory vs experiment for $1s$ energy shift	
	Problems	594
	References	596

AUTHOR INDEX	597
SUBJECT INDEX	602

Outline of Volume II

15	NON-ABELIAN GAUGE THEORIES
16	BACKGROUND FIELD METHODS
17	RENORMALIZATION GROUP METHODS
18	OPERATOR PRODUCT EXPANSIONS
19	SPONTANEOUSLY BROKEN GLOBAL SYMMETRIES
20	SPONTANEOUSLY BROKEN LOCAL SYMMETRIES
21	ANOMALIES
22	TOPOLOGICALLY COMPLICATED FIELDS
23	SUPERCONDUCTIVITY

Preface to Volume I

Why another book on quantum field theory? Today the student of quantum field theory can choose from among a score of excellent books, several of them quite up-to-date. Another book will be worth while only if it offers something new in content or perspective.

As to content, although this book contains a good amount of new material, I suppose the most distinctive thing about it is its generality; I have tried throughout to discuss matters in a context that is as general as possible. This is in part because quantum field theory has found applications far removed from the scene of its old successes, quantum electrodynamics, but even more because I think that this generality will help to keep the important points from being submerged in the technicalities of specific theories. Of course, specific examples are frequently used to illustrate general points, examples that are chosen from contemporary particle physics or nuclear physics as well as from quantum electrodynamics.

It is, however, the perspective of this book, rather than its content, that provided my chief motivation in writing it. I aim to present quantum field theory in a manner that will give the reader the clearest possible idea of *why* this theory takes the form it does, and why in this form it does such a good job of describing the real world.

The traditional approach, since the first papers of Heisenberg and Pauli on general quantum field theory, has been to take the existence of fields for granted, relying for justification on our experience with electromagnetism, and ‘quantize’ them — that is, apply to various simple field theories the rules of canonical quantization or path integration. Some of this traditional approach will be found here in the historical introduction presented in Chapter 1. This is certainly a way of getting rapidly into the subject, but it seems to me that it leaves the reflective reader with too many unanswered questions. Why should we believe in the rules of canonical quantization or path integration? Why should we adopt the simple field equations and Lagrangians that are found in the literature? For that matter, why have fields at all? It does not seem satisfactory to me to appeal to experience; after all, our purpose in theoretical physics is not just to describe the world as we find it, but to explain — in terms of a few fundamental principles — why the world is the way it is.

The point of view of this book is that quantum field theory is the way it is because (aside from theories like string theory that have an infinite number of particle types) it is the only way to reconcile the principles of quantum mechanics (including the cluster decomposition property) with those of special relativity. This is a point of view I have held for many years, but it is also one that has become newly appropriate. We have learned in recent years to think of our successful quantum field theories, including quantum electrodynamics, as ‘effective field theories,’ low-energy approximations to a deeper theory that may not even be a field theory, but something different like a string theory. On this basis, the reason that quantum field theories describe physics at accessible energies is that *any* relativistic quantum theory will look at sufficiently low energy like a quantum field theory. It is therefore important to understand the rationale for quantum field theory in terms of the principles of relativity and quantum mechanics. Also, we think differently now about some of the problems of quantum field theories, such as non-renormalizability and ‘triviality,’ that used to bother us when we thought of these theories as truly fundamental, and the discussions here will reflect these changes. This is intended to be a book on quantum field theory for the era of effective field theories.

The most immediate and certain consequences of relativity and quantum mechanics are the

properties of particle states, so here particles come first — they are introduced in Chapter 2 as ingredients in the representation of the inhomogeneous Lorentz group in the Hilbert space of quantum mechanics. Chapter 3 provides a framework for addressing the fundamental dynamical question: given a state that in the distant past looks like a certain collection of free particles, what will it look like in the future? Knowing the generator of time-translations, the Hamiltonian, we can answer this question through the perturbative expansion for the array of transition amplitudes known as the S -matrix. In Chapter 4 the principle of cluster decomposition is invoked to describe how the generator of time-translations, the Hamiltonian, is to be constructed from creation and annihilation operators. Then in Chapter 5 we return to Lorentz invariance, and show that it requires these creation and annihilation operators to be grouped together in causal quantum fields. As a spin-off, we deduce the CPT theorem and the connection between spin and statistics. The formalism is used in Chapter 6 to derive the Feynman rules for calculating the S -matrix.

It is not until Chapter 7 that we come to Lagrangians and the canonical formalism. The rationale here for introducing them is not that they have proved useful elsewhere in physics (never a very satisfying explanation) but rather that this formalism makes it easy to choose interaction Hamiltonians for which the S -matrix satisfies various assumed symmetries. In particular, the Lorentz invariance of the Lagrangian density ensures the existence of a set of ten operators that satisfy the algebra of the Poincaré group and, as we show in Chapter 3, this is the key condition that we need to prove the Lorentz invariance of the S -matrix. Quantum electrodynamics finally appears in Chapter 8. Path integration is introduced in Chapter 9, and used to justify some of the hand-waving in Chapter 8 regarding the Feynman rules for quantum electrodynamics. This is a somewhat later introduction of path integrals than is fashionable these days, but it seems to me that although path integration is by far the best way of rapidly deriving Feynman rules from a given Lagrangian, it rather obscures the quantum mechanical reasons underlying these calculations.

Volume I concludes with a series of chapters, 10–14, that provide an introduction to the calculation of radiative corrections, involving loop graphs, in general field theories. Here too the arrangement is a bit unusual; we start with a chapter on non-perturbative methods, in part because the results we obtain help us to understand the necessity for field and mass renormalization, without regard to whether the theory contains infinities or not. Chapter 11 presents the classic one-loop calculations of quantum chromodynamics, both as an opportunity to explain useful calculational techniques (Feynman parameters, Wick rotation, dimensional and Pauli–Villars regularization), and also as a concrete example of renormalization in action. The experience gained in Chapter 11 is extended to all orders and general theories in Chapter 12, which also describes the modern view of non-renormalizability that is appropriate to effective field theories. Chapter 13 is a digression on the special problems raised by massless particles of low energy or parallel momenta. The Dirac equation for an electron in an external electromagnetic field, which historically appeared almost at the very start of relativistic quantum mechanics, is not seen here until Chapter 14, on bound state problems, because this equation should not be viewed (as Dirac did) as a relativistic version of the Schrödinger equation, but rather as an approximation to a true relativistic quantum theory, the quantum field theory of photons and electrons. This chapter ends with a treatment of the Lamb shift, bringing the confrontation of theory and experiment up to date.

The reader may feel that some of the topics treated here, especially in Chapter 3, could more properly have been left to textbooks on nuclear or elementary particle physics. So they might, but in

my experience these topics are usually either not covered or covered poorly, using specific dynamical models rather than the general principles of symmetry and quantum mechanics. I have met string theorists who have never heard of the relation between time-reversal invariance and final-state phase shifts, and nuclear theorists who do not understand why resonances are governed by the Breit–Wigner formula. So in the early chapters I have tried to err on the side of inclusion rather than exclusion.

Volume II will deal with the advances that have revived quantum field theory in recent years: non-Abelian gauge theories, the renormalization group, broken symmetries, anomalies, instantons, and so on.

I have tried to give citations both to the classic papers in the quantum theory of fields and to useful references on topics that are mentioned but not presented in detail in this book. I did not always know who was responsible for material presented here, and the mere absence of a citation should not be taken as a claim that the material presented here is original. But some of it is. I hope that I have improved on the original literature or standard textbook treatments in several places, as for instance in the proof that symmetry operators are either unitary or antiunitary; the discussion of superselection rules; the analysis of particle degeneracy associated with unconventional representations of inversions; the use of the cluster decomposition principle; the derivation of the reduction formula; the derivation of the external field approximation; and even the calculation of the Lamb shift.

I have also supplied problems for each chapter except the first. Some of these problems aim simply at providing exercise in the use of techniques described in the chapter; others are intended to suggest extensions of the results of the chapter to a wider class of theories.

In teaching quantum field theory, I have found that each of the two volumes of this book provides enough material for a one-year course for graduate students. I intended that this book should be accessible to students who are familiar with non-relativistic quantum mechanics and classical electrodynamics. I assume a basic knowledge of complex analysis and matrix algebra, but topics in group theory and topology are explained where they are introduced.

This is not a book for the student who wants immediately to begin calculating Feynman graphs in the standard model of weak, electromagnetic, and strong interactions. Nor is this a book for those who seek a higher level of mathematical rigor. Indeed, there are parts of this book whose lack of rigor will bring tears to the eyes of the mathematically inclined reader. Rather, I hope it will suit the physicists and physics students who want to understand *why* quantum field theory is the way it is, so that they will be ready for whatever new developments in physics may take us beyond our present understandings.

* * *

Much of the material in this book I learned from my interactions over the years with numerous other physicists, far too many to name here. But I must acknowledge my special intellectual debt to Sidney Coleman, and to my colleagues at the University of Texas: Arno Bohm, Luis Boya, Phil Candelas, Bryce DeWitt, Cecile DeWitt-Morette, Jacques Distler, Willy Fischler, Josh Feinberg, Joaquim Gomis, Vadim Kaplunovsky, Joe Polchinski, and Paul Shapiro. I owe thanks for help in the preparation of the historical introduction to Gerry Holton, Arthur Miller, and Sam Schweber. Thanks are also due to Alyce Wilson, who prepared the illustrations and typed the $\text{\LaTeX}$ input files

until I learned how to do it, and to Terry Riley for finding countless books and articles. I am grateful to Maureen Storey and Alison Woollatt of Cambridge University Press for helping to ready this book for publication, and especially to my editor, Rufus Neal, for his friendly good advice.

Austin, Texas
October, 1994

STEVEN WEINBERG

Notation

Editorial note. This edition preserves Weinberg’s mostly-plus metric and physical content while translating several recurring conventions into notation common in modern quantum field theory. The complete replacement policy is recorded in the project’s notation guide.

Latin indices $i, j, k, \dots$ generally run over the three spatial coordinates 1, 2, 3. Greek indices $\mu, \nu, \dots$ run over spacetime coordinates in the order 0, 1, 2, 3, with x^0 the time coordinate. Repeated indices are summed unless otherwise indicated.

The spacetime metric is mostly plus,

$$\eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1),$$

so an on-shell four-momentum satisfies $p^2 = -m^2$, and spacelike separations satisfy $(x - x')^2 \geq 0$. The d’Alembertian is

$$\square = \partial_\mu \partial^\mu = \nabla^2 - \partial_t^2.$$

The Levi–Civita tensor is totally antisymmetric, with $\varepsilon^{0123} = +1$.

Spatial three-vectors are bold, and a hat denotes a unit vector: $\hat{\mathbf{v}} = \mathbf{v}/|\mathbf{v}|$. A dot denotes a time derivative.

Dirac matrices obey

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}, \quad \gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3.$$

With this metric convention, $\beta = i\gamma^0$ and the Dirac adjoint is $\bar{u} = u^\dagger\beta$. The step function $\theta(s)$ equals 1 for $s > 0$ and 0 for $s < 0$.

Complex conjugation, transpose, and Hermitian conjugation are written A^* , $A^\top$, and $A^\dagger$, respectively. The identity is **1**. The abbreviations +H.c. and +c.c. indicate addition of the Hermitian conjugate and complex conjugate of the preceding terms.

States use Dirac notation: $|p, \sigma\rangle$ for one-particle states, $|\Omega\rangle$ for the vacuum, and $\langle\phi | \psi\rangle$ for inner products. Asymptotic states are labeled $|\alpha\rangle_{\text{in}}$ and $|\beta\rangle_{\text{out}}$. Positive- and negative-frequency field pieces are written $\psi^{(+)}$ and $\psi^{(-)}$, avoiding the original overloading of + and – for both field frequencies and asymptotic states.

The interaction Hamiltonian is denoted

$$H_I(t) = \int d^3x \mathcal{H}_I(x),$$

in place of the original bare V notation. Other operators are generally left unhatted when the context is unambiguous.

Except in Chapter 1, $\hbar = c = 1$. Throughout, $-e$ is the rationalized charge of the electron, so $\alpha = e^2/(4\pi) \simeq 1/137$. Parentheses at the end of quoted numerical data give the uncertainty in the final digits.